



Complete Genome Sequence of Bacteriophage Finny, Isolated from a *Microbacterium foliorum* Culture

Tiffany Lee,^a Michaela Aguirre,^a Shey Andrews,^a Kayla Bahr,^a Abigail Ballard,^a Matthew Bristerpostma,^a Faith Cox,^a Leah Dowell,^a David Kiker,^a Tiffany Lujan,^a Stacy Luka,^a Abbigail Ramirez,^a Rheaven Sandoval,^a Kenneth Underhill,^a Haze Murphy,^a Cecilia Cabrera,^a Dustin Edwards^a

^aDepartment of Biological Sciences, Tarleton State University, Stephenville, Texas, USA

ABSTRACT Actinobacteriophage Finny contains a circularly permuted 40,313-bp double-stranded DNA genome with 63 predicted protein-coding genes. Finny was directly isolated from a soil sample collected in New Braunfels, Texas, that was incubated with *Microbacterium foliorum* SEA B-24224. Finny is closely related to bacteriophages MCubed, Andromedas, ColaCorta, Eleri, and Sansa.

Characterization of actinobacteriophage genomes provides valuable information on virus evolution and genetic diversity, including conservation of shared genes within a population (1). We report the genome sequence of actinobacteriophage Finny (2), directly isolated from soil inside a chicken coop in New Braunfels, TX (global positioning system [GPS] coordinates 29.782774N, 98.128783W). Soil samples were washed with peptone-yeast extract-calcium (PYCa) liquid medium and filtered through 0.22- μ m filters to extract the bacteriophage. The filtered medium was plated using a softagar overlay method on PYCa agar with *Microbacterium foliorum* strain SEA B-24224 at 29°C for 24 h. Bacteriophage replication formed small and medium lytic plaques with turbid halo rings. Finny was purified by picking a single plaque and placing it in 100 μ l phage buffer (10 mM Tris [pH 7.5], 10 mM MgSO₄, 68 mM NaCl, 1 mM CaCl₂, 10% glycerol), followed by two consecutive serial dilutions. Negative-staining transmission electron microscopy showed that the isolated bacteriophage had siphoviral morphology, with an approximate tail length of 150 nm and capsid diameter of 65 nm.

The bacteriophage DNA was extracted by a modified zinc chloride precipitation method (3) that included proteinase K (20 mg/ml) treatment to deactivate the nucleases prior to potassium acetate precipitation. Genomic sequencing libraries were prepared using the NEBNext Ultra II kit (New England BioLabs, Ipswich, MA); the libraries were pooled and sequenced with an Illumina MiSeq instrument at the Pittsburgh Bacteriophage Institute (Pittsburgh, PA). Sequencing was performed to approximately 7,918-fold coverage from 2,253,035 total single-end 150-base read length (4). A single bacteriophage contig was assembled from sequence reads using Newbler 2.9, with default settings, and Consed v29.0 (5) was used to perform quality control on assembly for inclusiveness and precision. We determined that the virus contains a double-stranded DNA genome 40,313 base pairs long, with 62.1% GC content. There were no accumulations of read starts or substantial coverage variations, so the genome was determined to be circularly permuted. The beginning of the genome was chosen by comparison to similar bacteriophage genomes.

Whole-genome nucleotide alignment with BLASTn (<https://blast.ncbi.nlm.nih.gov/>) (6) showed similar nucleotide identity to the bacteriophages summarized in Table 1 (2, 7).

Along with manual inspection, genome annotation was performed using GLIMMER v3.02 (8) and GeneMark v2.5p (9, 10) for correction and refinement of start sites and

Citation Lee T, Aguirre M, Andrews S, Bahr K, Ballard A, Bristerpostma M, Cox F, Dowell L, Kiker D, Lujan T, Luka S, Ramirez A, Sandoval R, Underhill K, Murphy H, Cabrera C, Edwards D. 2019. Complete genome sequence of bacteriophage Finny, isolated from a *Microbacterium foliorum* culture. Microbiol Resour Announc 8:e01039-19. <https://doi.org/10.1128/MRA.01039-19>.

Editor John J. Dennehy, Queens College

Copyright © 2019 Lee et al. This is an open-access article distributed under the terms of the [Creative Commons Attribution 4.0 International license](https://creativecommons.org/licenses/by/4.0/).

Address correspondence to Dustin Edwards, dcedwards@tarleton.edu.

Received 23 August 2019

Accepted 30 August 2019

Published 3 October 2019

TABLE 1 Characteristics of similar bacteriophages with *M. foliorum* as a host^a

Name	GenBank accession no.	Genome size (bp)	GC content (%)	No. of ORFs ^b	No. of tRNAs	% identity to Finny
Finny	MK894432	40,313	62.1	63	1	
MCubed	MN096378	40,381	62.0	63	0	97.44
Andromedas	MH590606	40,494	62.0	63	1	96.86
Eleri	MG839027	40,366	62.0	63	0	96.82
ColaCorta	MH590604	40,494	62.0	64	1	96.80
Sansa	MH513982	40,306	61.8	62	1	96.41

^a Similarity defined as an identity of >95%.^b ORFs, open reading frames.

revisions utilizing Phamerator (<https://phamerator.org/>) (11), DNA Master v5.23.2 (<http://phagesdb.org/DNAMaster/>), and PECAAN (<https://discover.kbrinsgd.org>). We predicted that Finny contained 63 protein-coding genes. One tRNA gene was identified by tRNAscan-SE v2.0 (12) and positioned at nucleotides 28924 through 28857. Start codon usage was determined to be 80.95% AUG and 19.05% GUG. Putative functions for 25 of the 63 predicted protein-coding genes were assigned using HHpred v3.0beta (13, 14) and NCBI BLASTp (6). The Finny genome is organized with genes 1 through 26 transcribed rightwards and encoding terminase, lysin A, and virion assembly and structural proteins. Genes transcribed leftwards encode RecA-like DNA recombinase, AAA-ATPase, Cas4 family exonuclease, DNA polymerase I, DNA helicase, MazG-like nucleotide pyrophosphohydrolase, thymidylate kinase, glycosyltransferase, and ThyX thymidylate synthase.

Data availability. The genome sequence of actinobacteriophage Finny is available at GenBank under accession number [MK894432](#). The raw reads are available in the SRA under accession number [SRX6700907](#).

ACKNOWLEDGMENTS

Support for this research was provided by Tarleton State University College of Science and Technology and by the Howard Hughes Medical Institute Science Education Alliance-Phage Hunters Advancing Genomics and Evolutionary Science (SEA-PHAGES) research and education program.

We thank Graham Hatfull, Welkin Pope, Deborah Jacobs-Sera, Daniel Russell, Rebecca Garlena, and Phoebe Doss for their technical support during the imaging of the virion and the isolation, sequencing, and annotation of this genome.

REFERENCES

- Pope WH, Bowman CA, Russell DA, Jacobs-Sera D, Asai DJ, Cresawn SG, Jacobs WR, Jr, Hendrix RW, Lawrence JG, Hatfull GF, Science Education Alliance Phage Hunters Advancing Genomics and Evolutionary Science, Phage Hunters Integrating Research and Education, Mycobacterial Genetics Course. 2015. Whole genome comparison of a large collection of mycobacteriophages reveals a continuum of phage genetic diversity. *Elife* 4:e06416. <https://doi.org/10.7554/eLife.06416>.
- Russell DA, Hatfull GF. 2017. PhagesDB: the actinobacteriophage database. *Bioinformatics* 33:784–786. <https://doi.org/10.1093/bioinformatics/btw711>.
- Santos MA. 1991. An improved method for the small scale preparation of bacteriophage DNA based on phage precipitation by zinc chloride. *Nucleic Acids Res* 19:5442. <https://doi.org/10.1093/nar/19.19.5442>.
- Russell DA. 2018. Sequencing, assembling, and finishing complete bacteriophage genomes, p 109–125. *In* Clokie MRJ, Kropinski AM, Lavigne R (ed), *Bacteriophages: methods and protocols*, vol 3. Springer, New York, NY.
- Gordon D, Green P. 2013. Consed: a graphical editor for next-generation sequencing. *Bioinformatics* 29:2936–2937. <https://doi.org/10.1093/bioinformatics/btt515>.
- Altschul SF, Gish W, Miller W, Myers EW, Lipman DJ. 1990. Basic local alignment search tool. *J Mol Biol* 215:403–410. [https://doi.org/10.1016/S0022-2836\(05\)80360-2](https://doi.org/10.1016/S0022-2836(05)80360-2).
- Ali I, Jones AE, Mohamed A, Sivanathan V. 2018. Genome sequences of Ilzat and Eleri, two phages isolated using *Microbacterium foliorum* NRRL B-24224. *Genome Announc* 6:e00144-18. <https://doi.org/10.1128/genomeA.00144-18>.
- Delcher AL, Harmon D, Kasif S, White O, Salzberg SL. 1999. Improved microbial gene identification with GLIMMER. *Nucleic Acids Res* 27:4636–4641. <https://doi.org/10.1093/nar/27.23.4636>.
- Borodovsky M, Mills R, Besemer J, Lomsadze A. 2003. Prokaryotic gene prediction using GeneMark and GeneMark.hmm. *Curr Protoc Bioinformatics* Chapter 4:Unit 4.5. <https://doi.org/10.1002/0471250953.bi0405s01>.
- Besemer J, Borodovsky M. 2005. GeneMark: Web software for gene finding in prokaryotes, eukaryotes and viruses. *Nucleic Acids Res* 33:W451–W454. <https://doi.org/10.1093/nar/gki487>.
- Cresawn SG, Bogel M, Day N, Jacobs-Sera D, Hendrix RW, Hatfull GF. 2011. Phamerator: a bioinformatic tool for comparative bacteriophage genomics. *BMC Bioinformatics* 12:395. <https://doi.org/10.1186/1471-2105-12-395>.
- Lowe TM, Chan PP. 2016. tRNAscan-SE On-line: integrating search and context for analysis of transfer RNA genes. *Nucleic Acids Res* 44:W54–W57. <https://doi.org/10.1093/nar/gkw413>.
- Söding J, Biegert A, Lupas AN. 2005. The HHpred interactive server for protein homology detection and structure prediction. *Nucleic Acids Res* 33:W244–W248. <https://doi.org/10.1093/nar/gki408>.
- Zimmermann L, Stephens A, Nam S-Z, Rau D, Kübler J, Lozajic M, Gabler F, Söding J, Lupas AN, Alva V. 2018. A completely reimplemented MPI Bioinformatics Toolkit with a new HHpred server at its core. *J Mol Biol* 430:2237–2243. <https://doi.org/10.1016/j.jmb.2017.12.007>.